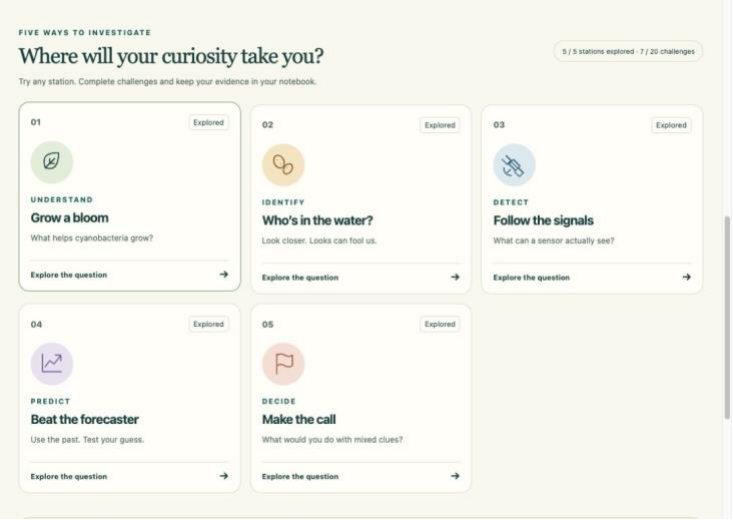
Lake Lab Teacher Guide

Facilitation notes and user instructions for middle school learners

Use this guide to teach the current Lady Bird Lake edition of Lake Lab. The aim is evidence literacy: students compare observations, test simplified models, identify uncertainty and decide what they need to learn next. The app supports a team's own project; it does not supply a finished FLL solution.



Students may enter any station. A station visit and a completed challenge are different progress measures.

What students should be able to do

- Distinguish real observations, derived summaries and simulations.
- Make a controlled comparison and save its settings.
- Explain why a sensor, photo or forecast can miss a local mat hazard.
- Compare model errors and justify a decision under uncertainty.

Pages	Teaching support
2–3	Setup, classroom routines and session plans
4	Local science and evidence boundaries
5–10	Station walkthroughs and expected observations
11–12	Assessment, support and troubleshooting
13	Source and image credits

This guide matches the October 10, 2026 local review build. Do not use the app as a live lake-safety advisory.

Prepare the app and the room

Setup

1. Open the app on each type of device you will use. Select a station, change a slider and check that charts and images load.

2. For local use, run python3 -m http.server 8765 --bind 127.0.0.1 inside the lake-lab folder, then open http://127.0.0.1:8765/ on that computer. Do not open index.html directly.

3. For student devices, supply the live link: https://chintan2108.github.io/lake-lab/ .

4. Check that students can save a note, reload it and download a printable copy. Managed school browsers may block storage or downloads.

5. Explain that shared browser profiles share the same notebook. Use separate profiles/devices where possible, or agree who exports the group's work.

6. Keep the student guide available. Ask students to use the same device and browser for return visits.

Privacy and storage routine

There are no student accounts, analytics or live external-data requests. Notes and progress use local browser storage, with an in-memory fallback. A fallback warning means work may vanish when the page closes. The site host can still have normal access logs.

Ask for observations without student names or personal details. Export at the end of each session. The current app has no JSON import control or cross-device sync. Clearing browser data or Reset local progress can remove work; export first.

How the interface fits together

Explore opens the station cards. Each station has Try it, Take a challenge, How this works and For your project. The challenge check saves completion; it does not assess a student's explanation. Station notebook saves include settings. Direct notebook entries are general reflections.

Keyboard operation uses native controls: Tab for focus, arrows for sliders and menus, Enter or Space for actions. Offer the tabular data alongside visual graphs. Avoid grading speed or small visual differences.

Choose a teaching sequence

A 30 minute introduction

Time	Activity	Evidence to collect
0–5 min	Introduce bottom mats and real versus simulated badges.	One reason a clear surface can mislead.
5–15 min	Understand: a cool/warm comparison with other controls fixed.	Two recorded settings and a written comparison.
15–24 min	Detect: compare 10 m and about 1 km pixels.	One feature that a coarse pixel could miss.
24–30 min	Notebook reflection and export.	One claim, one limit and one next question.

A 60 minute workshop

Use 5 minutes for the local context, 12 for Understand, 12 for one Detect tool, 15 for Predict, 8 for a short Decide discussion and 8 for notebook synthesis. Treat this as a sample route, not a race to complete every challenge.

Three sessions for deeper inquiry

- Session 1: Understand and Identify. Compare controlled model changes with the uncertainty of images.
- Session 2: Detect and Predict. Ask what is measured, what is missing and how to test on new examples.
- Session 3: Decide and notebook synthesis. Debate consequences and define a team-designed next investigation.

A routine that transfers across stations

Before a click, ask for a prediction and reason. After a click, ask for an observation. Then ask what the evidence does not support. Students can rotate the roles of operator, evidence recorder and questioner.

End with: “What would make us change our minds?” Preserve competing explanations. Do not let a check mark or a high game score substitute for the team’s own reasoning.

Teach the local science accurately

Lady Bird Lake is a narrow, dammed reach of the Colorado River between Tom Miller and Longhorn dams. Its harmful-cyanobacteria concern since 2019 has prominently involved benthic mats, which can detach and float. The app's floating-bloom mode teaches a general contrast; it is not the dominant local-hazard model.

A growth-support score is not toxin probability. An organism identification is not a toxin assay. A pigment signal is not a direct toxin measurement. Clear water and missing reports do not establish safety. Keep these distinctions visible in student explanations.

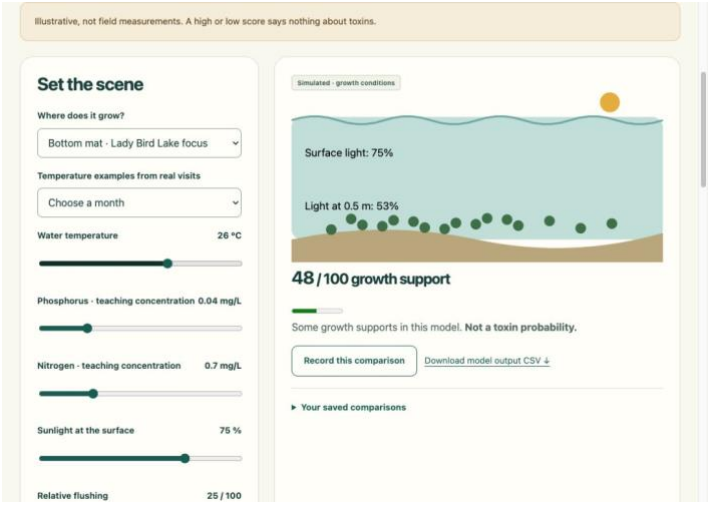
App evidence	Boundary to teach
City water observations	Discrete shallow-water visits at selected sites. Provisional U/J flags are retained but not authoritatively decoded.
Monthly temperature examples	Uneven visit medians, not a complete climatology. January, April and December are absent; some months have only three site-days.
Toxin evidence	Selected 2019 samples with differing media, analytes and units. Mat ng/g and water µg/L are not interchangeable.
Flow and weather	USGS flow is downstream of Longhorn Dam. NOAA air temperature and rain are at Camp Mabry, not in the lake.
Images and maps	Historical images and geometry, not current conditions. The grid origin is a teaching choice.
Simulations	Author-chosen rules, not a calibrated local growth or toxin forecast.

Handle qualifiers and missingness

Below-limit counts reflect explicit less-than qualifiers. "Unqualified" means a source field lacks that qualifier; it does not by itself confirm a verified positive. Missingness may reflect sampling decisions, season or source availability. Do not manufacture binary safe/unsafe labels from these records.

The frozen environmental window is 2016–2025. Real-data Predict mode explores evidence and limitations; no real-toxin classifier is trained. Consult the app's Data & sources and methods files for exact screening rules.

Facilitate Understand



Use a fixed set of other controls to separate the effect of one change.

1. Open Understand and Reset scene. Record 15°C, then 28°C, leaving other settings unchanged.

2. Ask students to explain the difference using “in this teaching model”. Compare mat and floating modes at identical settings.

3. Use the challenge setup buttons before the October and one-change tasks. Require a short notebook explanation after the check passes.

Expected observations and challenge support

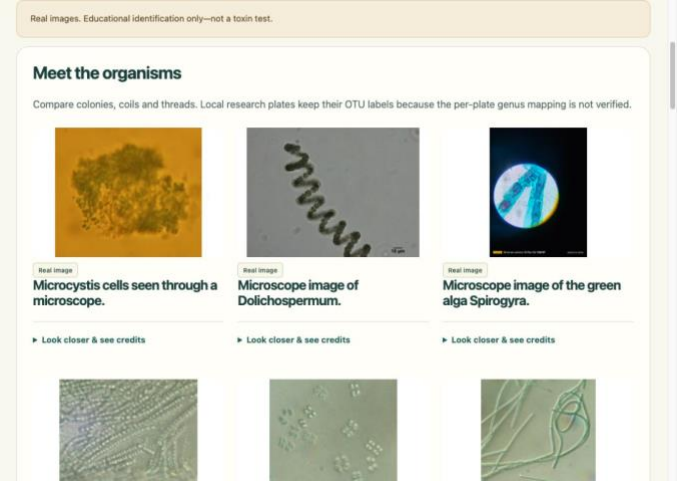
At default nutrients, sunlight and flushing in mat mode, the model gives about 27/100 at 15°C and 49/100 at 28°C. Its preferred temperature is 28°C, so increasing temperature beyond that does not continually raise growth support. Scores describe relative support, not biomass or risk.

Follow the warmth: record one setting at ≤15°C and one at ≥25°C, with all other controls equal. October: Load October sets temperature to 25°C from a rounded observed median; keeping it fixed and raising phosphorus to 0.2 and nitrogen to 3 gives about 58/100 with the other defaults. These nutrient settings are teaching values, not a lake recommendation.

Change just one thing: use Start one-change scene, then change only relative flushing from 5 to 100; the score drops by more than 20. Bottom comparison: record both modes at the same settings. The mat receives less light because a fixed 0.5 m bottom and attenuation are assumed.

Ask: Which omitted process could change this result? Could the same growth score occur with different combinations? Do not imply that the sliders are actions the students should carry out in the lake.

Facilitate Identify



Real microscope images support close observation, not toxin diagnosis. Image authors and licences are listed on page 13.

1. Have students open Look closer & see credits on three cards. Ask for observations before names.

2. In Teach the machine, label at least three training originals and run Train nearest-neighbour model.

3. Change one label and retrain. Compare predictions, nearest neighbours and feature distances.

4. Complete the field-photo quiz and distinguish visible identity from invisible toxin status.

What the model can teach

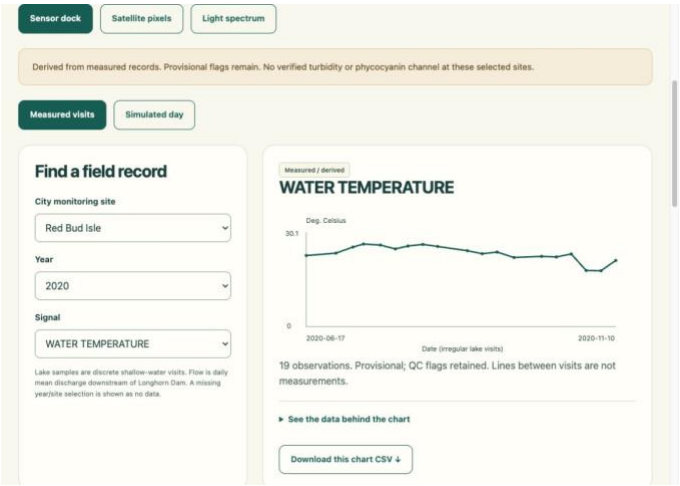
This is 1-nearest-neighbour classification using mean RGB brightness and horizontal edge contrast from a 32 × 32 image. Features can reflect lighting, background or plate layout instead of biology. There is no pretrained neural network and no student-image upload.

All four training originals are cyanobacteria examples. The two held-out originals are Microcystis and Spirogyra. Labelling all training images cyanobacteria produces one correct result out of two. This is an intentionally inadequate training collection, not a usable two-class recognizer. Each test answer changes accuracy by 50 percentage points.

A wrong label can accidentally improve this tiny test score. That is evidence against trusting the score, not permission to relabel biology. Additional independent, verified examples of both classes would be needed for a meaningful classifier exercise. Local OTU-to-genus mapping is still unresolved.

Field-quiz key: the published Red Bud Isle figure shows mats; duckweed is a look-alike; the toxin question requires More evidence needed. A photo alone establishes neither safety nor toxicity. Challenge checks require three opened cards, at least three labels and training, a changed label and retraining, and the three correct field answers.

Facilitate the sensor dock



The displayed temperature record is selected by site, year and parameter. It is not continuous monitoring.

1. Open Detect → Sensor dock → Measured visits. Select Red Bud Isle, 2020 and WATER TEMPERATURE.

2. Ask students to read units and sample dates before describing a trend. Open the underlying table.

3. Switch to PH, DISSOLVED OXYGEN or downstream flow. Explain the different spatial and temporal support.

4. Switch to Simulated day. Compare pH, oxygen and phycocyanin before answering the timing question.

Explain each channel

Temperature measures warmth. pH describes acid–base conditions. Dissolved oxygen is oxygen available in water. Turbidity describes cloudiness from suspended material and is not a cyanobacteria-specific signal. A phycocyanin probe measures a pigment fluorescence response; calibration and sampling position matter.

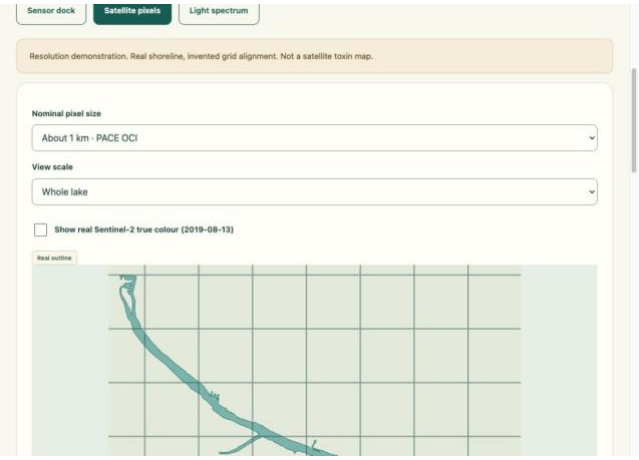
No verified local phycocyanin time series is included. The selected-site snapshot has no turbidity channel. Those daily curves are clearly simulated. RFU cannot be treated as µg/L without an instrument-specific relationship.

Discussion and challenge support

The invented daily pH and dissolved-oxygen curves rise before the pigment pulse. Select “pH and dissolved oxygen” and Check this comparison, then check the challenge. Do not teach that this temporal order holds in every real lake.

For the first challenge, visit both measured and simulated views. Ask students to identify their different evidence status. Blank charts are legitimate source gaps, not zero values. A downstream discharge record cannot locate lake currents or a shoreline mat.

Facilitate satellite and spectrum comparisons



A nominal 1 km grid over a narrow lake demonstrates mixed land–water cells. It is not a satellite toxin product.

Satellite pixels

Have students compare 10, 300 and 1,000 m at a fixed view scale; use 4× for fine grid lines. Toggle the dated Sentinel-2 image if helpful. The challenge requires viewing 10 and 1,000 m. Ask what the same pixel might include besides water.

Sentinel-2 visible bands can be 10 m, but it has no narrow 620 nm band. Sentinel-3 OLCI is nominally 300 m; PACE OCI is about 1 km at nadir. These displayed sizes are instructional. Actual footprints, alignment and usable lake retrievals require further checks. The app does not promise a resolved CyAN time series for Lady Bird Lake.

Light spectrum

1. Select Light spectrum. Keep chlorophyll-a at 10 µg/L; record PC = 0.

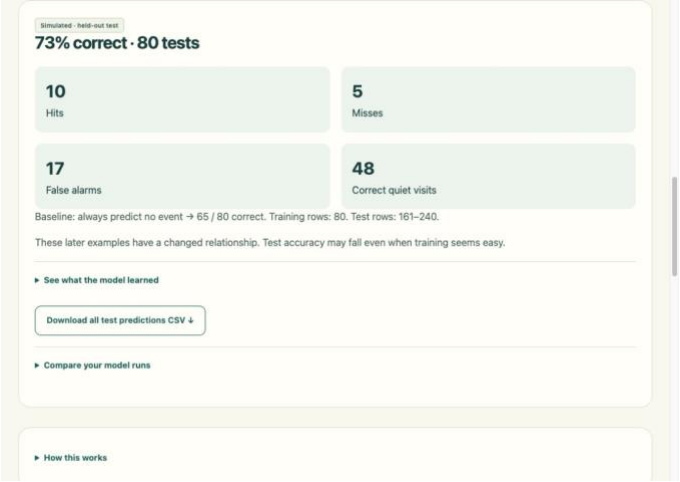
2. Raise PC to 30 µg/L or more, record again and inspect values near 620 nm.

3. Explain absorption, backscattering and Rrs as a relative reflected-light measure; students need not derive the equations.

PC raises absorption near 620 nm while chlorophyll is fixed. This is not toxin absorption. The challenge checks a zero-PC and ≥30-PC pair at equal chlorophyll.

The forward equations and PC term come from Water Optics Lab. Analytic teaching shapes replace numerical reference tables whose reuse rights were not verified. Numerical spectra therefore differ from the sibling. Bottom reflectance, atmosphere and sensor band integration are omitted. See docs/optical-methods.md for every coefficient and limitation.

Facilitate Predict



One model scores 73% correct, yet its always-no-event baseline gets 65 of 80 correct. Ask which errors matter.

1. Start in Simulated training game. Ask students to commit to a guess before revealing an outcome.

2. Choose temperature and nutrient inputs, set 80 training examples and train. Compare with the baseline.

3. Change only the inputs or training count. Retrain and inspect Compare your model runs.

4. Open Real evidence explorer, reveal the next sample groups and discuss why they are not clean binary labels.

Interpret the result instead of rewarding the percentage

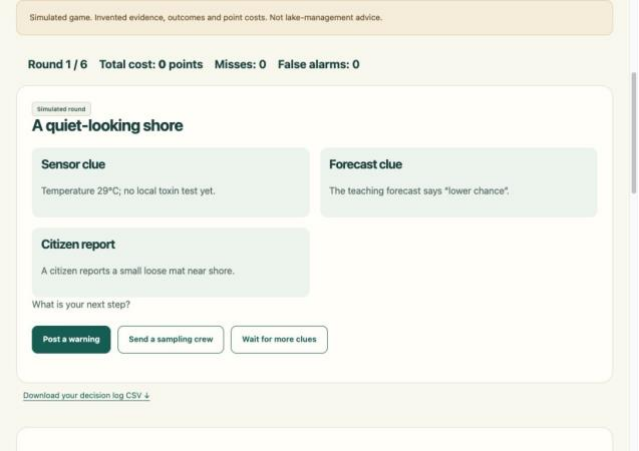
There are 240 invented visits. Training uses the first 40, 80, 120 or 160; the last 80 are always held out. Scaling and logistic-regression fitting use training rows only. A later nutrient relationship changes, illustrating distribution shift. More inputs or history need not improve results.

With temperature and nutrient inputs and 80 training examples, the captured result has 10 hits, 5 misses, 17 false alarms and 48 correct quiet visits: 58/80, rounded to 73%. The no-event baseline gets 65/80 but misses all 15 events. Neither accuracy alone nor a single error count defines the best decision.

The model weights combine normalized clues and a 0.5 cutoff creates an event/no-event prediction. They are associations in invented data, not real causal effects. Repeated tuning on the same held-out set weakens its independence as a final evaluation.

Challenge support: make a guess; train once; train two distinct input/history combinations; reveal real records. Also require a verbal or notebook statement of uncertainty—the app's check alone does not assess that reasoning.

Facilitate Decide



Ask for a reason before revealing the outcome. The citizen evidence is a written report in this game.

Action	Event	No event
Warn	2	5
Wait	12	0
Sample then act	3 plus the chosen action cost	3 plus the chosen action cost

These are teaching points, not monetary valuations or official policy. Lower total cost is better only under these chosen rules. Sampling instantly reveals the hidden result in the game; real sampling has delay, spatial gaps and uncertainty.

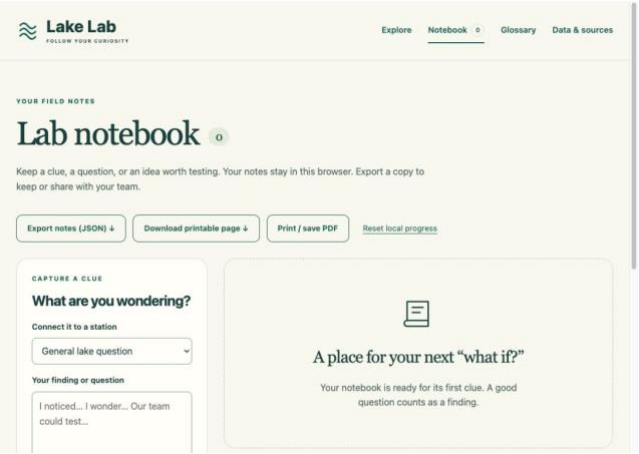
Facilitation sequence

Read all three clues. Ask what supports each action and what is missing. If the team samples, require a second decision after the test. After the reveal, distinguish a poor outcome from an unreasonable decision made with limited evidence. Finish six rounds and export before Replay, which clears the current decision log.

Teacher reference: hidden outcomes are event, no event, event, event, no event, no event. Always warning costs 21 points; always waiting costs 36. Sampling each time and choosing after the reveal costs 24. Do not provide these totals before the first run. They illustrate this game, not a preferred real-world policy.

Challenge support: finish one round; sample and then decide; finish all six; replay and make a new decision. Ask whose interests the cost rules capture and whose they omit. A memorized second run is not evidence of a reliable forecasting strategy.

Use the notebook for assessment



Station notes preserve settings; general notebook notes are written reflections. Use the printable export for readable student work.

Collect reasoning rather than completion alone

Look for	Evidence of understanding
Status of evidence	Names the record as measured, derived or simulated.
Controlled comparison	States what changed, what stayed fixed and what was observed.
Interpretation	Connects the observation to a limited claim, without turning it into safety advice.
Uncertainty and next test	Names a missing factor and proposes evidence that could change the claim.

Use these four dimensions for formative feedback rather than a speed leaderboard. A challenge check confirms an activity condition; it is not a rubric for scientific quality. Students can revise a note after a discussion.

Example: “In the mat model, I changed temperature from 15°C to 28°C. I held nutrients, light and flushing fixed. Growth support increased. This simulation cannot tell us whether a real mat contains toxins. We would need different evidence for that question.”

At the end, export the printable HTML and the JSON copy; collect only what your classroom process requires. There is no import button or cloud sync. CSV chart downloads are separate from the notebook. Keep simulation labels and source credits when students use outputs in judging materials.

Support students and resolve common problems

Adjust the task without removing the science

- For younger readers, start with one slider or one pixel comparison. Read the badge and project prompt together.
- Pair a student who operates the controls with a student who describes the evidence; switch roles.
- Use the data tables, browser zoom and keyboard controls. Do not require colour alone to explain a result.
- For extension, compare baselines, identify sampling bias, inspect coefficients or propose an independent validation design.
- For a short online meeting, model one comparison and a notebook save; leave other stations as choices.

Problem	What to check
The lab will not open	Serve the folder over HTTP. Copy all relative data/assets. Double-clicking index.html is insufficient.
A chart has no points	Check site/year/parameter. Try the documented 2020 temperature example. Preserve genuine gaps.
A challenge does not pass	Use its setup control, meet exact comparison conditions and record before checking.
Training gives nothing	Select at least one forecast input or image label; the image challenge requires at least three labels.
Old results or settings appear	The same browser retains progress. Reset only after exporting and with the student’s agreement.
Notes disappear	Check blocked/cleared storage, private mode, device or site-address changes. Export before leaving.
Print does not open	Download printable page, open the HTML in a normal browser and print/save PDF there.
Numbers differ from a screenshot	Check settings and model version. Screenshot progress is demonstration state; it is not a required result.

Current boundaries: no live safety status, no verified local phycocyanin series, no real-toxin training, no validated species classifier and no fully audited accessibility certification. The home screen may still carry an early-edition label; this guide covers the five implemented stations.

Sources and credits for teaching

For detailed provenance, open Data & sources → Methods, limits & downloads. The local repository also contains data/derived/README.md, docs/methods.md, docs/optical-methods.md and docs/review/PHASES-2-5.md. Review scientific uncertainties before presenting model output as evidence about the actual lake.

Source	Reference
City of Austin	austintexas.gov/watershed-protection/algae-austins-waterways ; data.austintexas.gov/d/5tye-7ray
USGS gauge 08158000	waterdata.usgs.gov/monitoring-location/USGS-08158000/
NOAA weather	GHCN-Daily station USW00013958, Austin Camp Mabry; see frozen metadata.
Local study	Perri et al. 2024, Toxins 16, 91. doi.org/10.3390/toxins16020091
Local image dataset	data.mendeley.com/datasets/f94329byhk/2
Optical sibling	chintan2108.github.io/Water_Optics_Rrs_Simulator/

Lake Lab was created by Chintan B. Maniyar. Guide text and original app illustrations: CC BY 4.0. Screenshots show the local app on October 10, 2026. Some counters and settings reflect demonstration use. Your screen may differ. Third-party images keep their own licences.

Microscope images visible in the Identify screenshot: Microcystis by Panek, CC BY 2.5; Dolichospermum by Masa Zupancic, CC BY-SA 4.0; Spirogyra by SPJ777, CC BY 4.0. Local OTU plates: Katherine Perri, Schonna Manning, Matt Ashworth and Brent Bellinger, CC BY 4.0. Images are shown within a screenshot and resized for the page; labels and colours were not retouched.

Image sources: commons.wikimedia.org/wiki/File:Microcystis_live.jpg ; commons.wikimedia.org/wiki/File:Dolichospermum_sp..jpg ; commons.wikimedia.org/wiki/File:Microscopic_image_of_spirogyra.jpg ; data.mendeley.com/datasets/f94329byhk/2 . Licence terms: creativecommons.org/licenses/by/2.5/ ; creativecommons.org/licenses/by/4.0/ ; creativecommons.org/licenses/by-sa/4.0/ .

The pixel screenshot uses USGS NHD geometry, public domain. The optional satellite layer in the app contains modified Copernicus Sentinel data 2019, processed by ESA and hosted by Element 84 / AWS; it is not displayed in this guide's grid screenshot.

Creator contact: chintanmaniyar@gmail.com · linkedin.com/in/chintanmaniyar/ . Related project: cyanotracker.uga.edu . Repository target: github.com/Chintan2108/lake-lab . These external references are not live data connections inside the app.